Web GIS for Planning Future Shelter Locations in Seattle

Sponsor Organization: SHARE/WHEEL

Miles Winarske, Aditya Ayappa, Ellie Velez

University of Washington, Department of Geography

GEOG 469 SPR 25: Geographic Information Systems Workshop

June 2025

Executive Summary

This project results from the Geography 469 Senior Capstone course during the spring quarter of 2025. Students were assigned a sponsor organization with an issue they needed help solving through the professional development skills taught through the geography data science major track. For this project, our group worked with the organization known as SHARE: Seattle's Housing and Resource Effort. They reached out looking for a mapping tool that would allow them to locate future shelter site locations easily, provide details regarding essential features of a given future shelter site, and then qualify each site based on their needed criteria.

In an effort to provide a future-proof tool, we saw a web-based GIS application as a means of solving their problem. We approached the situation on multiple fronts: front-end User interface, Backend locally-hosted database, website hosting, and data science methods. We also provided the GitHub repository with an open-source license, which allows many different organizations that may be interested in this application to use the framework code and redevelop it as they see fit. This allows more flexibility within the community, without risking the security of SHARE's site directory.

Front-end development was conducted through HTML, CSS, and JavaScript scripting. We used the open-source Mapbox GL API for our mapping interactivity, providing useful interactive features. CSS was used to provide styling standardization, and HTML was used for website navigation & indexing. The only issue we encountered when developing our User Interface was keeping the same aesthetic styling across the various pages on the website.

For the backend development, we used the Flask API Python package to exchange.GeoJSON and communicate to a development server that feeds to HTTP addressing. The Data Science aspect of Python allows us to save.GeoJSON data is locally in the Project GitHub repository and is updated through the website-based user

commits. The GeoPandas Python package was used to process POIs, specifically transit access, and public facilities like hospitals and schools. For website hosting, we have employed the services of Render.com, a free hosting service that allows for full-stack web service hosting. Website updating is coordinated between render.com and the project's GitHub repository. Initially, we attempted to use Netlify, a similar yet purely static free website hosting service. The issue we faced with netlify.com was that as we progressed through our development from front-end to full stack, we determined that static website hosting would not allow for a database, especially one at low to no cost for upkeep. The shift from Netlify.com to Render.com allowed us to save site locations created by users and maintain data to ensure it works properly throughout the codebase.

Table of Contents

1. Sponsor, Background, and Problem Statement..... 1-2
1.1 About the Sponsor..... 1
1.2. Project Background.....2
1.3. Problem Statement.....2
2. Data Integration Strategies.....3-5
2.1 Computing transit access for land parcels in Seattle..... 3
2.2. User Interface..... 4
2.3. Spatial Feature Buffers..... 5
3. Project Digital environments..... 6-7
3.1. JavaScript.....6
3.2. Python.....6
3.3. HTML.....6
3.4. CSS.....6
3.5. Render.com.....6
3.6. Wordpress.com..... 7
3.7. ArcGIS Pro.....7
3.8. GitHub.....7
3.9. Google Drive..... 7
4. Data Acquisition.....8
5. Findings and Discussion of Results.....9
6. Summary, Conclusions, and Recommendations 10-11
6.1. Section Summary..... 10
6.2. Proposed Solution..... 10
6.3. Recommendations.....10-11
7. References.....12-13
8. Appendix..... 14
Appendix A - Contact List..... 14
Appendix B - Project Resources.....14-15
Appendix C - Project Team Resumes.....15

1. Sponsor, Background, and Problem Statement

1.1 About the Sponsor

Our sponsor for this project was the *Seattle Housing and Resource Effort*, more commonly known as *SHARE*. Established in 1990, SHARE is a grassroots, self-managed organization founded by the unhoused and provides support & shelter for people experiencing homelessness. As an organization, its mission is to create safe, dignified shelter and provide essential resources to unhoused individuals across the Seattle area.

SHARE is one of the largest shelter networks in King County, where it operates both indoor shelters and tent cities. The indoor shelters include men-only, women-only, and co-ed facilities. As for the tent cities, they are democratically run encampments that typically house up to 100 Co-ed residents. These encampments often reside within church properties around Seattle and the Eastside and are supported by faith-based and community partners. Despite their mobility, these communities maintain a highly organized structure and enforce a strict Code of Conduct focused on sobriety, nonviolence, and shared responsibility.

In partnership with its sister organization, WHEEL (Women's Housing, Equality, and Enhancement League), SHARE provides nightly shelter for over 350 people. Together, they operate critical support programs such as the SHARE2 Housing-for-Work initiative, a storage locker program, and public hygiene centers. These services complement SHARE's shelter efforts by addressing unhoused individuals' broader challenges, offering practical resources alongside a pathway to stability.

1.2. Project Background

For this project, we were specifically asked to create a digital mapping tool that allows staff members of SHARE/WHEEL to locate, delegate, and leave notes about future indoor and outdoor shelter locations. The goal of this project was to transition from having a stack of loose notes to having a coordinated, live-service GIS tool that is easily accessible to staff members. The staff recognized that security was a significant need for their organization, so having a login prompt to prevent other organizations from using their information to poach shelter sites was vital.

1.3. Problem Statement

SHARE needs a digital tool to coordinate shelter site locations and determine which characteristics best suit their future shelters.

SHARE suggested having a feature to see nearby facilities, such as hospitals or Schools, and transit access to public transportation (Bus Stations, Train Stations, etc.). The purpose of this is to recognize the existing community infrastructure, how well the parcel will support the Organization, and to prevent potential issues when communicating with local stakeholders.

2. Data Integration Strategies

2.1 Computing transit access for land parcels in Seattle

Computing transit access for land parcels in Seattle involves analyzing how many bus stops are within a walkable distance, typically 800 meters, of each parcel. This data assigns a transit score to each parcel, indicating how well it is served by public transportation. On the map, these scores are visualized using a color-coded layer, allowing users to quickly see which areas have excellent, moderate, or poor access to transit. An appropriate color classification scheme was chosen to reflect the distribution of transit scores across the dataset. This helps inform site selection and planning decisions, especially when overlaid with other layers like shelter locations. Figure 1 is an example of a small section from the map:

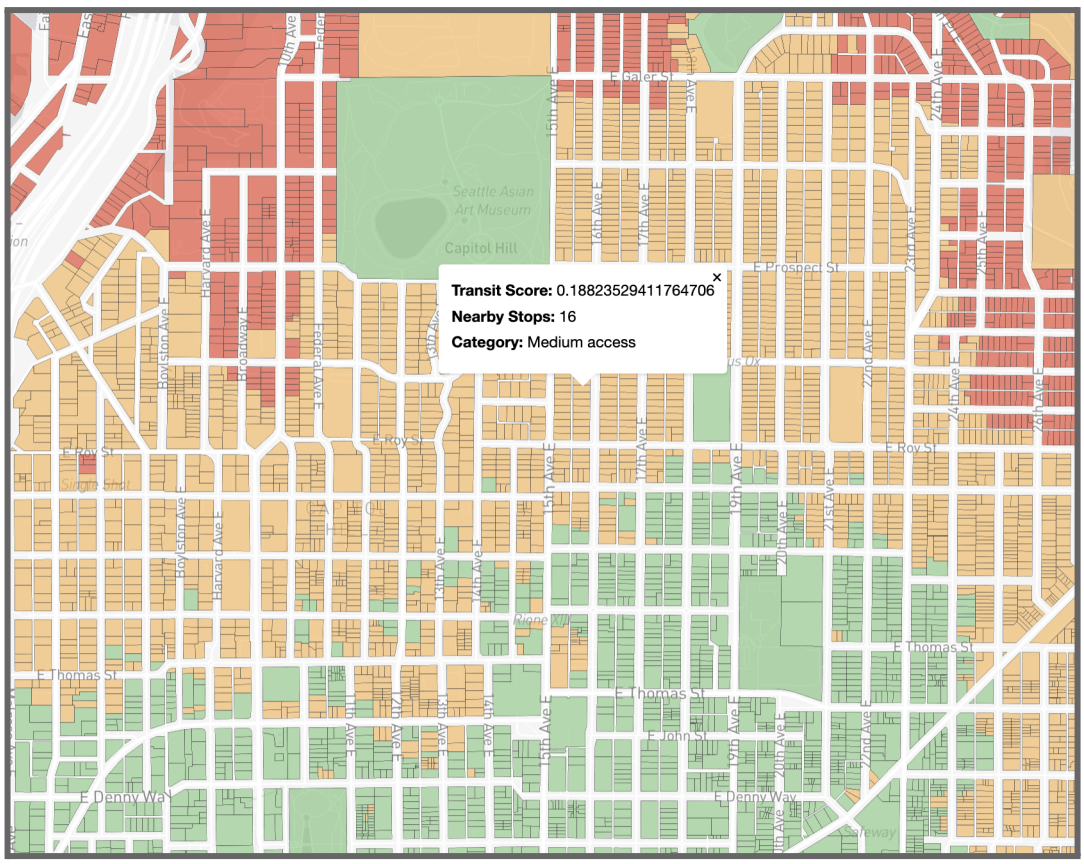

Figure 1. *Transit Accessibility Analysis*

2.2. User Interface

The interactive shelter site allows users to search through existing shelters, view detailed information, and add new shelters through a form. This makes it easy for planners to manage and contribute to the shelter dataset without manually editing files. The backend is powered by a lightweight Flask server, which handles form submissions and updates the GeoJSON file in real time. When a new shelter is submitted, Flask appends the latest data to the existing GeoJSON, ensuring that updates are persistent and reflect on the map immediately. The images below show the process of adding a new shelter and how those changes interact with the backend.

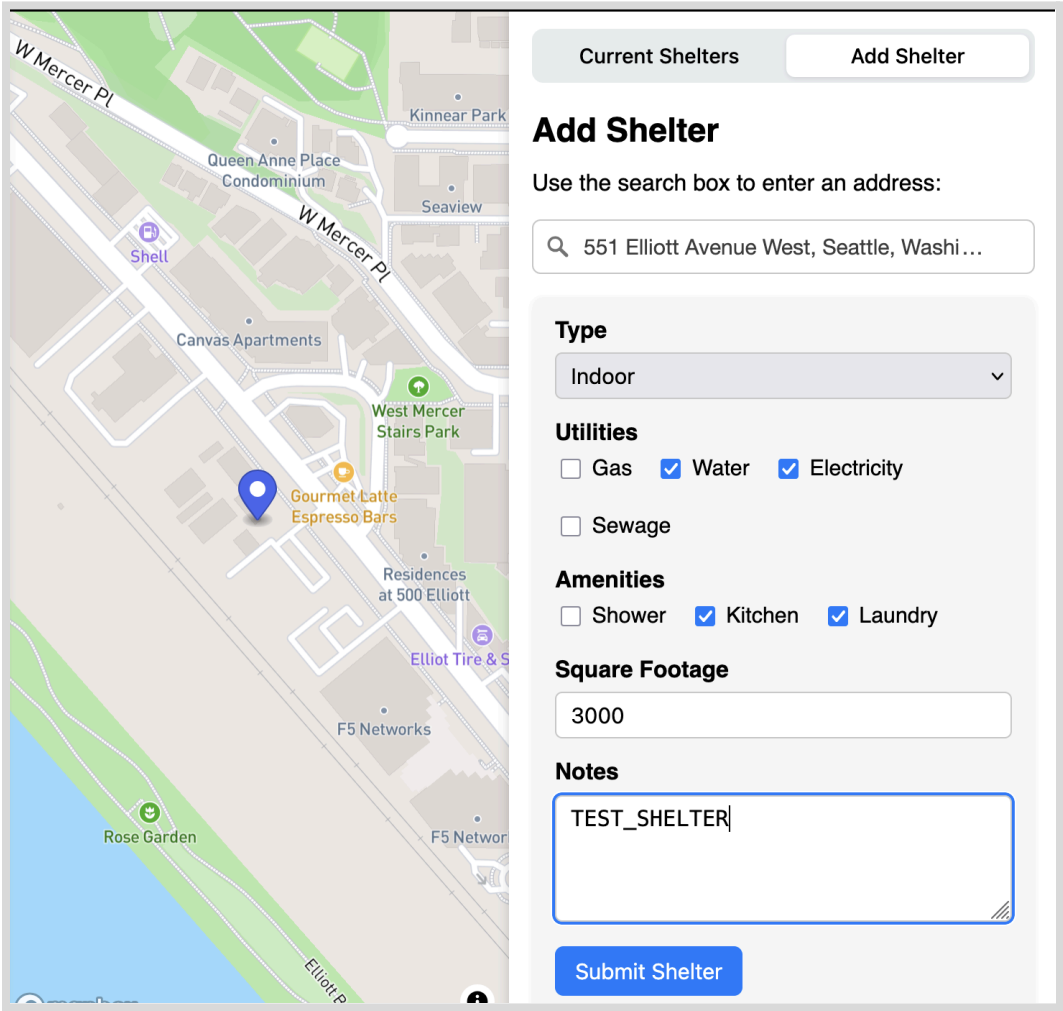

Figure 2. *Example of Adding a Shelter Site Location*

```
    "properties": {
      "name": "551 Elliott Avenue West, Seattle, Washington 98119, United States",
      "shelter_type": "indoor",
      "gas": false,
      "water": true,
      "electricity": true,
      "sewage": false,
      "amenities": [
        "kitchen",
        "laundry"
      ],
      "notes": "TEST_SHELTER"
    }
  }
```

Figure 3. *Codebase for Site Dictionary in Python 3*

2.3. Spatial Feature Buffers

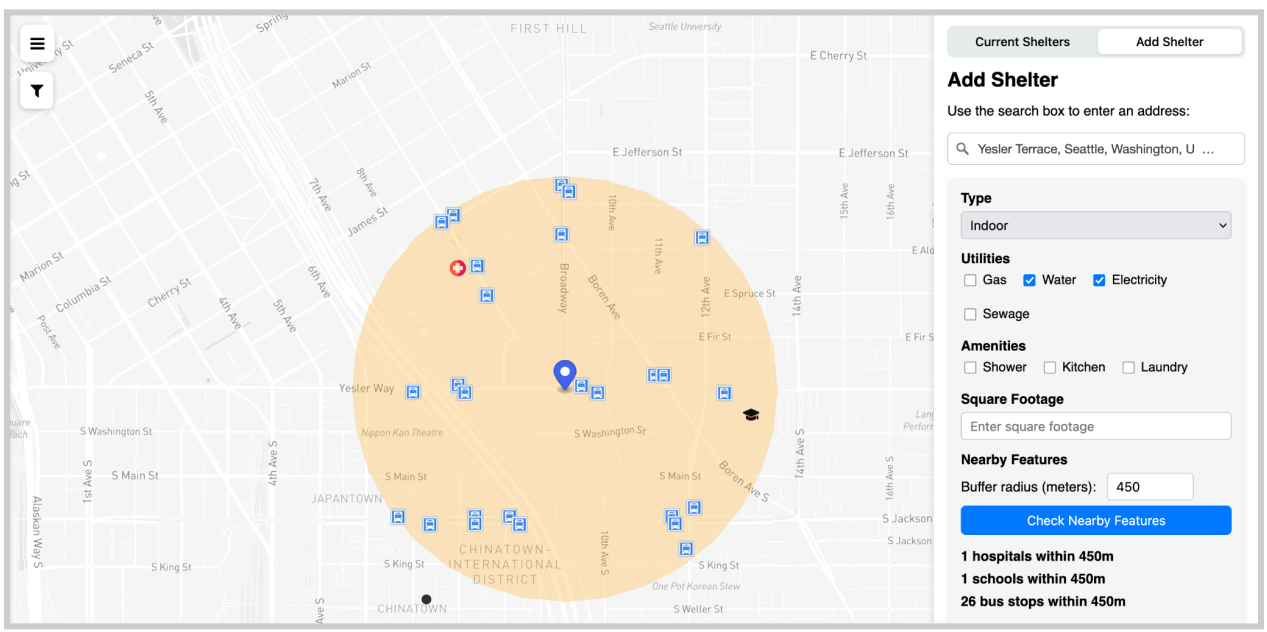

Figure 4. *Spatial Buffer Plotting Accessibility*

The feature shown allows users to add new shelter locations and assess their suitability. Users can search an address and choose a buffer radius on the map (e.g., 450 meters), highlighting nearby features such as hospitals, schools, and bus stops , which is essential for accessibility and service planning. This functionality helps organizations make better decisions by visualizing key infrastructure and services.

3. Project Digital environments

For this project, we used full-stack development with multiple coding languages to create a user interface for deep interactivity and communication between a local database. We used GitHub for Version Control, Render.com for Hosting, WordPress.com for a Custom Domain, and ArcGIS Pro for data analysis and cleaning data.

3.1. JavaScript

JavaScript was used to add responsive user interfaces and interactive map elements. Specifically, we used Mapbox GL JS, a client-side JavaScript library, to create the web map, enabling us to display it on the browser. This API allows us to add user interactivity and customize our website's map experience.

3.2. Python

The Flask API in Python was utilized to allow users to interact with a server over HTTP and exchange data in GeoJSON format. Our web application enables newly added shelter data to be stored and updated locally in a GeoJSON file. The Geopandas library was used to process points of interest, such as transit access around potential shelter sites

3.3. HTML

HTML was used to define and structure the contents of the website.

3.4. CSS

CSS was used to design the layout and appearance of the website, and helped keep the styling consistent between web pages.

3.5. Render.com

Hosting service used to deploy a full-stack web service, where deployment of updates is coordinated through GitHub commits.

3.6. Wordpress.com

Custom Domain licensed to us at an 11.99/year lease for 5 years. Our group funds this to provide a fully cohesive website that is easy to access from any web browser and for any person.

3.7. ArcGIS Pro

ArcGIS Pro was used to clip the spatial boundary of the parcel dataset so that we would select only parcels in Seattle and remove the rest of the data. This was done primarily to shrink the data file size and clear away unnecessary information, which would slow down the upload and render times.

3.8. GitHub

We used a shared repository for version control across team devices, which includes a readme.txt.

3.9. Google Drive

To coordinate group efforts within the group regarding written rhetoric, we utilized a shared Google Drive. Within the drive, we have literature for our academic process, including interview questions, a Gantt chart detailing the project timeline, the Project Scope, important information about the project sponsor & key literature surrounding the project context, the initial progress report for the course, Team protocol for production, an initial website mockup to guide design efforts, and the Website writing detailing the User Guide instructions and the about page.

4. Data Acquisition

Data for this Project was sourced from three different sources: King County's GIS OpenData Portal, the City of Seattle's GIS OpenData Portal, and Sound Transit's Open Transit Data. The aim of using these three public, governed sources was to ensure that the data collection methods had a level of standardization between them, therefore supporting the connection between location data points. The data we used consisted of:

a. **Public Schools in Seattle**
    i. Spatial location of Public Schools that may be in proximity to shelter site locations. The purpose of this dataset was to advise SHARE members of where schools are in proximity to shelter site locations, and provide an Icon to show spatial proximity.

b. Hospitals in Seattle
    i. Spatial location of publicly managed hospitals in proximity to the shelter site. The purpose of this dataset was to advise SHARE members of where Hospitals are in proximity to shelter site locations, and provide an Icon to show spatial proximity. A key issue addressed by Michele Marchand was the redirection of hospital staff to high-risk of non-sobriety homeless individuals to the nearest shelter. The key is to avoid a site location where this would occur, because SHARE/WHEEL shelters require a thorough screening effort.

c. Parcels for King County with the Property Information/parcel address area
    i. Provides the necessary information about each property within the Seattle municipal boundaries, including the square footage of each parcel, the owner of each parcel, and land use type of each parcel.

d. **King County Metro GTFS data for bus routes and stops**
    i. This dataset allows for spatial proximity analysis in tandem with future shelter site locations. The purpose of this dataset was to advise SHARE members where bus/train stations are in proximity to shelter site locations and provide an icon to show spatial proximity.

5. Findings and Discussion of Results

The final product streamlines the process of adding and managing new shelter locations in Seattle. The intuitive UI includes a right-side panel where users can input shelter details such as name, address, and utilities, which are then updated to the site's GeoJSON data.

The collapsible panel on the left allows the user to filter through the various shelters displayed both on the map and on the right panel. In addition to data input, the map interface allows users to analyze spatial relationships between shelter locations and important features such as transit access, hospitals, and schools.

A noteworthy feature layer was added later in the project workflow: a Transit Accessibility layer. The Access layer provides a visual choropleth map, visualizing the availability of transit stops within a range of aggregated parcel blocks.

On adding a new shelter location, the user can select a buffer around the shelter within which various points of interest are plotted on the map. This visualisation allows users to assess the spatial suitability of a new shelter site. In addition to visually inspecting the site's suitability relative to its surrounding features, a summary of the key statistics are provided as well.

6. Summary, Conclusions, and Recommendations

6.1. Section Summary

This section will outline how our group connected the problem statement with the proposed solution and implementation strategies, as well as a discussion of proposed future work to continue progress on this project.

6.2. Proposed Solution

To address the need for a digital tool to assist SHARE's Site location efforts, our team developed a free, open-source GIS mapping application designed to assist in the location, evaluation, and documentation of potential shelter sites across Seattle. We have provided numerous features to assist in recognizing amenities, features, and positive aspects of each address within the Map database, which will assist in discovering key locations for the best shelter sites possible. The tool is interactive, filterable, and updatable, therefore empowering SHARE with greater spatial decision-making capabilities.

6.3. Recommendations

As a means of making this project open-source and low-cost, our team used free-to-operate resources. Our main goal was to create a codebase that is easily modifiable when a new feature is requested. Because our resources for hosting the website and database are free, there are some minor limitations for high usage by a large group of individuals.

1) Allocation of more resources: As a tool that is vital to SHARE's efforts, it is important that it is constantly updated, and this requires a collaborative effort between the Project team and SHARE staff. As per the final meeting for this course, SHARE and the Project team have formally agreed to continue collaborating to continue work on this project with open communication channels. This may be between Email and Phone communication, with the project agreement as a basis for understanding between the two parties.

2) Single Sign-on: This will be hosted through SupaBase: a free-to-use database software that allows projects to have a cloud-based data service with User Login functionality and Coordination between website deployment and Database updating. SupaBase will provide the project with the capability to seamlessly integrate login and new shelter site information in a timely manner, and will ensure that minimal data loss occurs.

3) Expanding and Updating the Data: Information constantly changing across time and although ours is up to date, allowing the site to extract the latest data would result in more accurate spatial analysis for new shelter sites. Furthermore, enriching our dataset by collecting more features such as blackouts, natural lighting , and walkability scores would result in a more robust platform, further refining decision making.

4) Adding LLM support: Integrating a Large Language Model (LLM) into the platform can significantly enhance user experience. Users could interact with the system using natural language queries, asking specific questions like “which locations have the best transit access and least noise pollution”? This would enable better interpretability amongst complex geospatial data. In addition, the LLM can assist in data entry and validation. For example, when adding new shelter site information, the LLM can automatically check for missing fields, summarize key site attributes, and even suggest tags or categories based on context.

5) A final recommendation, as the course-related work on the project ceases, is to find partners in the municipal system with more resources related to property management and ownership. This will help propel the project efforts regarding the availability of indoor and outdoor space for shelter sites, and can help SHARE staff determine the correct contact for the building ownership.

7. References

Brallier, S., & and Southworth, S. (n.d.). Rational choice? Choosing the street over the shelter. *Journal of Social Distress and Homelessness*, *0*(0), 1–9. https://doi.org/10.1080/10530789.2024.2379162

Exploratory analysis of the homeless shelter system in columbus, ohio: Geografiska Annaler: Series B, Human Geography: Vol 87, No 1. (n.d.). Retrieved June 10, 2025, from https://www.tandfonline.com/doi/abs/10.1111/j.0435-3684.2005.00178.x

Hospitals. (n.d.). Retrieved May 27, 2025, from https://data-seattlecitygis.opendata.arcgis.com/datasets/hospitals-2/about

Mosher-Ashley, P. M., & and Henrikson, N. (1997). Shelter-Based Services in Massachusetts for Homeless Adult Individuals. *Social Work in Health Care*, *26*(2), 15–32. https://doi.org/10.1300/J010v26n02_02

mtwinarske/SHARE-Shelter-Site-Map: An interactive map for the Organization SHARE/WHEEL, based out of Seattle, WA. This map has capabilities for adding new locations, and has a multitude of interactivity features. (n.d.). Retrieved May 27, 2025, from https://github.com/mtwinarske/SHARE-Shelter-Site-Map/tree/main

Open Transit Data (OTD) | OTD downloads | Sound Transit. (n.d.). Retrieved May 27, 2025, from https://www.soundtransit.org/help-contacts/business-information/open-transit-data-otd/otd-downloads

O'Toole, T. P., Gibbon, J. L., Hanusa, B. H., & Fine, M. J. (1999). Preferences for sites of care among urban homeless and housed poor adults. *Journal of General Internal Medicine*, *14*(10), 599–605. https://doi.org/10.1046/j.1525-1497.1999.09258.x

Parcels for King County with Address with Property Information / parcel address area. (n.d.). Retrieved May 27, 2025, from https://gis-kingcounty.opendata.arcgis.com/datasets/kingcounty::parcels-for-king-county-with-address-with-property-information-parcel-address-area/about

Seattle Public Schools Sites 2023-2024. (n.d.). Retrieved May 27, 2025, from https://data-seattlecitygis.opendata.arcgis.com/datasets/SeattleCityGIS::seattle-public-schools-sites-2023-2024/about

Transportation needs and mobility patterns of persons experiencing homelessness following shelter decentralization—ScienceDirect. (n.d.). Retrieved June 10, 2025, from https://www-sciencedirect-com.offcampus.lib.washington.edu/science/article/pii/S0149718923000836?via%3Dihub

Wilking, J., Madonia, G., & Hansen, P. (2025). Analysis of crime around a low-barrier, rotating homeless shelter. *Socio-Economic Planning Sciences*, *98*, 102173. https://doi.org/10.1016/j.seps.2025.102173

8. Appendix

Appendix A - Contact List

Alphabetical, First/Last

Aditya Ayappa
Project Member
(206) 532-7658
Aditya8@uw.edu

Ellie Velez
Project Member
(206) 972-5830
gnvelez@uw.edu

Miles Winarske
Project Member
(206) 532-7658
mtwin@uw.edu

Michele Marchand (SHARE/WHEEL)
Share Organizer
(206) 956-0334
Shelters@sharewheel.org

Appendix B - Project *Resources*

1. Sample GTFS Feed – King County Metro Transit
Sound Transit Open Transit Data
General Transit Feed Specification (GTFS) datasets containing spatial information for regional transit routes and stops. Used in proximity analysis of bus stops to potential shelter sites.

2. Seattle Parcel Dataset with Property Information

King County GIS Open Data Portal
Parcel shapefile with ownership, zoning, and land use details. Used to define shelter site locations and assess suitability based on parcel characteristics.

3. Seattle Public Schools Location Data
Seattle GIS Open Data Portal
Spatial dataset of public school locations used to assess proximity of proposed shelter sites to schools.

4. Hospital Locations – City of Seattle
Seattle GIS Open Data
Used to map and analyze hospital access for each proposed shelter site.

5. Literature Sample
Brallier, S., & Southworth, S. Journal of Social Distress and Homelessness.
Academic article referenced in contextual analysis of shelter use preferences and housing vulnerability.

6. Project GitHub Repository
Project Repository
Repository contains the full codebase and data files used in the web application. Referenced to coordinate data flows and feature implementation.

7. User Guide and Project Documentation
Internal Project Google Drive Repository
Internal planning documents such as the project scope, team protocols, interview notes, and visual mockups that guided development.

Appendix C - Project *Team Resumes*

Project Members:

1. Aditya Ayappa Resume
2. Ellie Velez Resume
3. Miles Winarske